

Collections



Data Types

- **Exact Numerics**
- **Approximate Numerics**
- **Date and Time**
- **Character Strings**
- **Binary Strings**
- **Unicode Character Strings**
- **Other Data Types (XML,SQLvariant,GUID)**

Associative Entities

Mapping Associative Entities

- Identifier Not Assigned
 - Default primary key for the association relation is composed of the primary keys of the two entities (as in M:N relationship)
- Identifier Assigned
 - It is natural and familiar to end-users
 - Default identifier may not be unique

Figure 5-15: Mapping an associative entity

(a) Associative entity

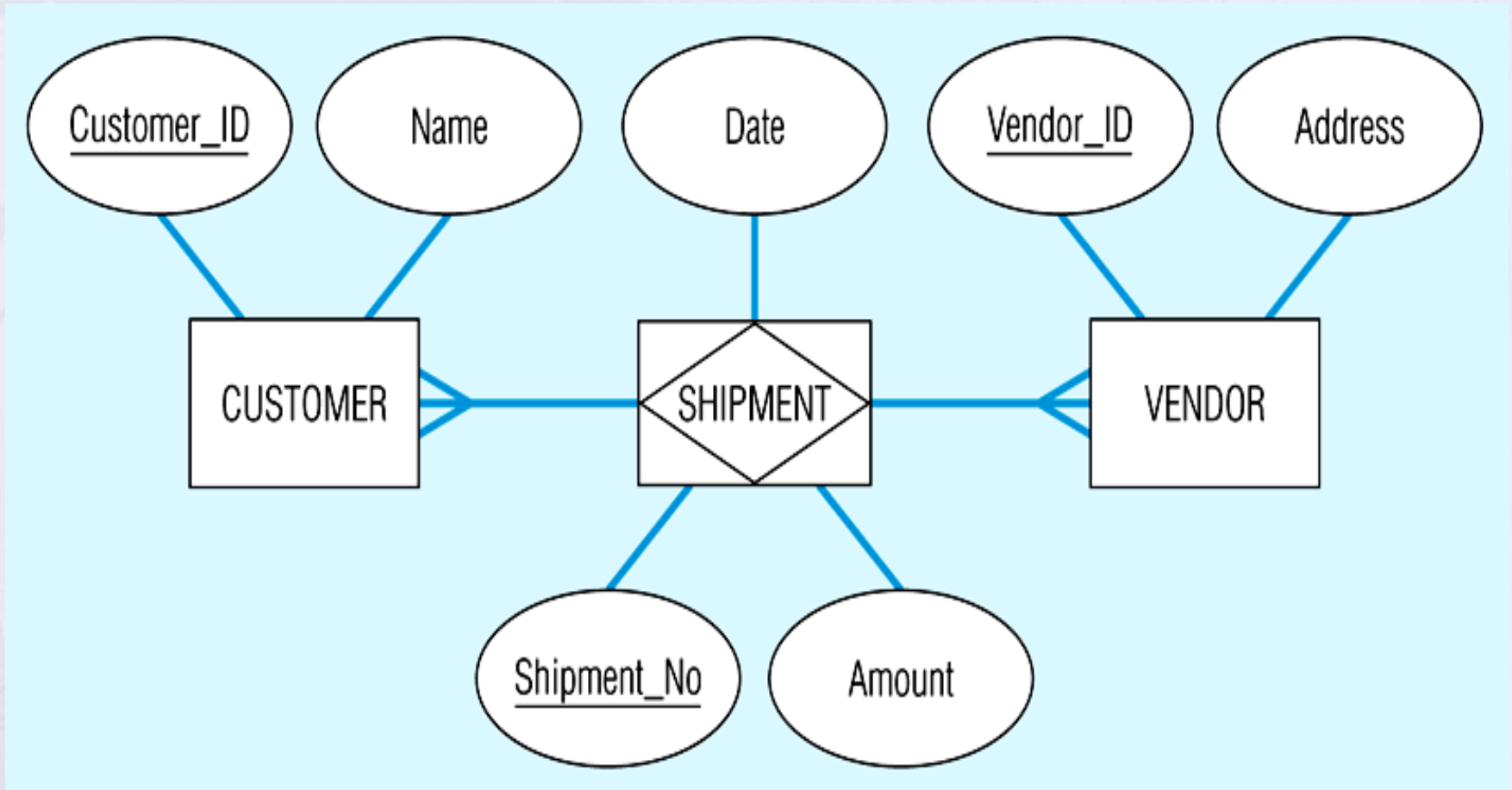


Figure 5-15(b) Three resulting relations

CUSTOMER

<u>Customer_ID</u>	Name	(Other Attributes)
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SHIPMENT

<u>Shipment_No</u>	<u>Customer_ID</u>	<u>Vendor_ID</u>	Date	Amount
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VENDOR

<u>Vendor_ID</u>	Address	(Other Attributes)
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SQL Server Versions History

1st Generation

SQL Server Version	Features
SQL 6.0/6.5 (1995)	First version designed specifically for Windows NT Replication
SQL Server 4.2 (1992)	Developed for Windows NT 3.1
SQL Server 1.0 (1989)	Developed by Microsoft, Sybase, and Ashton-Tate for OS/2

2nd Generation

SQL Server Version	Features
SQL2000	Focus on Performance and Scalability XML support Data Mining Reporting Services
SQL Server 7.0 (1999)	Restructure of Relational Server Data Transformation Services Online Analytical Processing

SQL Server Versions History

3 rd Generation	
SQL Server Version	Features
SQL 2012,2014	Always On Power View File Table Sequence Data Quality Service
SQL2008/SQL2008 R2	Power Pivot Enhance SharePoint Integration Enhance T-SQL (Ranking, Merge, Output) Improve and enhance for BI Tools
SQL2005	High Availability(includes Database Mirroring) Security Enhancements (DB Schema) Integration Services SQLCLR XML and Web services supports

SQL Server 2012 Editions

Edition	Description
Enterprise	For large scale, business-critical applications
Standard	For small/medium, departmental applications
BI Edition	For BI Services
Express	Entry level/learning edition
Web	For low-end Web hosting

Database Objectives

Objects	Notes
<i>Tables</i>	Contain all the data in SQL Server databases
<i>Views</i>	Act like a virtual table or a stored query
<i>Indexes</i>	Enable fast retrieval, built from one or more columns in table or view
<i>Triggers</i>	Execute a batch of SQL code when an insert, update or delete command is executed against a specific table
<i>Procedures</i>	Accept parameters, contain statements, and return values.
<i>Constraints</i>	Prevent inconsistent data from being placed in a column
<i>Rules</i>	Specify acceptable values that can be inserted in column

Definition

➤ **Data Warehouse:**

- A subject-oriented, integrated, time-variant, non-updatable collection of data used in support of management decision-making processes
- ***Subject-oriented:*** e.g. customers, patients, students, products
- ***Integrated:*** Consistent naming conventions, formats, encoding structures; from multiple data sources
- ***Time-variant:*** Can study trends and changes
- ***Nonupdatable:*** Read-only, periodically refreshed